# Pegasus himoyalangan texnik va notexnik bayoni

PEGASUS

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Hujjat turi: patent uslubidagi nazoratli texnik dossye. Versiya: bir nuqta bir. Oshkor qilish holati: arxitektura darajasidagi bayon, implementatsiya qo'llanmasi emas. Hujjat Pegasus logistika tizimi nima qilishi, tizim nima uchun farqli ekani, asosiy control loop'lar qanday ishlashi va ixtironi qaysi matematik landshaftlar shakllantirishini tasvirlaydi. Hujjat ataylab source code, private endpoint map'lar, schema internals, private thresholds, deployment quantities, credential flows va model constants'ni oshkor qilmaydi.

## TAVSIF

Pegasus supplier'lar, factory operator'lari, warehouse operator'lari, driver'lar, retailer'lar va payload team'lar uchun ko'p rolli logistika operating system hisoblanadi. Asosiy g'oya oddiy aytiladi. Logistika biznesiga bitta order, bitta truck, bitta manifest yoki bitta payment haqida bahslashadigan oltita uzilgan application kerak emas. Pegasus operatsiyani shared state network sifatida ko'radi. Har bir rol o'ziga tegishli network qismini ko'radi, har bir muhim action governed transition orqali o'tadi, va system automation'ni foydali saqlab, accountability'ni yo'qotishiga yo'l qo'ymaydi.

Texnik ixtiro role-scoped interface'lar, durable event propagation, geospatial decisioning, manifest control, financial reconciliation va exception governance'ni bitta coherent operating model ichida bog'lashdan iborat. Notexnik ixtiro esa responsibility intizomida. Supplier commercial policy belgilaydi, factory replenishment tayyorlaydi, warehouse dispatch'ni boshqaradi, driver verified delivery bajaradi, retailer demand va receipt'ni tasdiqlaydi, payload operator esa loading integrity'ni himoya qiladi. System bu mas'uliyatlarni xiralashtirmaydi. Ularni birlashtiradi.

Hujjat joriy repository architecture va qo'llab-quvvatlovchi documentation asosida yozilgan. Implementatsiya Go backend services, chi routing, Spanner-backed transactional truth, Redis cache va Pub/Sub invalidation, Kafka eventing through transactional outbox, WebSocket role channels, Next.js web portals, native Android va iOS apps hamda payload terminal'dan foydalanadi. Bu concrete technology'lar disclosure'ni asoslash uchun tilga olinadi, lekin himoyalanayotgan ixtiro bitta vendor yoki framework'dan kengroq. Ixtiro distributed logistics execution'ni scale, partial failure va human override sharoitida consistent saqlaydigan control model'dir.

## ASOS

Ko'p logistics software biror narsa buzilmaguncha integratsiyalashgandek ko'rinadi. Route o'zgaradi, warehouse short bo'ladi, truck unavailable bo'ladi, payment reconciliation talab qiladi yoki driver system kutganidan oldin location'ga yetadi. O'sha paytda weak design ko'rinadi. Inventory bir narsa deydi. Dispatch boshqa narsa deydi. Driver app stale work ko'rsatadi. Portal request two hundred response qaytargani uchun success deydi. Finance mismatch'ni keyin topadi.

Oddiy fix — ko'proq dashboard. Bu odamlarga muammoni ko'rishga yordam beradi, lekin muammoni olib tashlamaydi. Qiyinroq problem presentation emas. Qiyinroq problem state authority. Modern logistics platform qaysi actor qaysi object'ni o'zgartira olishini, qaysi boshqa role'lar xabardor qilinishi kerakligini, qaysi financial consequence kelishini, qaysi cached view'lar invalidated bo'lishini, qaysi automation pause bo'lishini va nima bo'lganini qaysi audit trail isbotlashini bilishi kerak.

Pegasus bunga har bir important workflow'ni governed state transition sifatida ko'rish orqali javob beradi. System supplier, factory yoki warehouse scope uchun client request body'dagi friendly field'ga tayanmaydi. Scope authenticated role context'dan chiqariladi. System event delivery'ni database write'dan keyingi umidli side effect deb ko'rmaydi. Durable mutation va durable propagation birga bog'lanadi. Manual intervention automated dispatch bilan race qilishiga yo'l qo'yilmaydi. Lock semantics insonga machine bilan urishmasdan plan'ni override qilish imkonini beradi.

Notexnik background ham muhim. Operator'lar microservice haqida o'ylamaydi. Ular consequence haqida o'ylaydi. Bu truck'ni yuborish safe'mi. Bu manifest sealed'mi. Retailer delivery kutyaptimi. Payment clean'mi. Warehouse act qilishga allowed'mi. Foydali platform bu human questions'ni precise technical constraints'ga aylantirishi, ammo human'ga implementation burden yuklamasligi kerak.

## CHIZMALARNING QISQACHA TAVSIFI

Drawing set repository architecture map, machine-readable architecture graph, technology inventory, existing visual asset library, Mermaid workflow diagrams va backend route-composition graph asosida quriladi. References professional patent drawings uchun descriptive evidence sifatida ishlatiladi. Ular public implementation map emas va source code, private endpoints, schema internals, production topology, secrets, thresholds yoki model constants'ni oshkor qilmaydi.

Infrastructure drawing tashqaridan ichkarigacha complete operating envelope'ni ko'rsatishi kerak. Web portals, desktop shells, native mobile clients va payload terminal Maglev-style routing and protection boundary orqali kiradi. Shu boundary ortida chi router backend domain handlers'ga dispatch qiladi, WebSocket hubs live role channels'ni saqlaydi, Spanner transactional truth'ni ushlab turadi, Redis cache va Pub/Sub invalidation beradi, Kafka outbox path orqali durable events oladi, worker services automation and reconciliation bajaradi, observability esa activity'ni traceable system evidence'ga bog'laydi.

Ingress and request-routing drawing client'lar sticky sessions yoki local server memory'ga emas, stateless service layer'ga kirishini ko'rsatishi kerak. Route-composition graph supplier, retailer, driver, payload, factory, warehouse, order, delivery, payment, telemetry, webhook, proximity, catalog, simulation, treasury va infrastructure families uchun grouped contract surfaces sifatida berilishi kerak. Bu drawing platform scattered endpoint fragments bilan emas, role and domain contracts bilan tashkil qilinganini ko'rsatadi.

Role responsibility drawing supplier'ni commercial and operational owner, factory role'ni production-side readiness, warehouse role'ni local fulfillment control, driver role'ni physical execution, retailer role'ni demand and receipt confirmation, payload role'ni loading integrity sifatida ko'rsatishi kerak. Drawing bu role'lar free-form data sharing bilan emas, governed handoff'lar orqali ulanganini ham ko'rsatadi.

Role-surface drawing har bir role isolated app emas, product row ekanini ko'rsatishi kerak. Supplier, retailer, driver, payload, factory va warehouse backend contracts, applicable web yoki desktop surfaces, applicable Android clients, applicable iOS clients, applicable terminal surfaces va realtime channels bilan bog'lanadi. Bu drawing parity across role row system property ekanini, cosmetic delivery preference emasligini aniqlashtirishi kerak.

Ecosystem workflow drawing full business loop'ni ko'rsatishi kerak. Demand and planning supplier policy, catalog, pricing, zones, factories, warehouses va retailer order intent bilan boshlanadi. Fulfillment warehouse stock pressure, factory supply response, payload loading, manifest sealing, driver execution, retailer receipt confirmation, exception handling, analytics, replenishment va next demand cycle orqali o'tadi. Existing workflow diagrams va generated SVG visuals drawing uchun basis beradi.

Governed transition drawing business action system'ga permission emas, intent sifatida kirishini ko'rsatadi. System authenticated identity'ni resolve qiladi, supplier yoki node scope'ni derive qiladi, policy'ni checks qiladi, requested state transition'ni evaluate qiladi, stale yoki replayed attempts'ni rejects qiladi, state change'ni commit qiladi, durable downstream evidence emits qiladi, affected read models'ni invalidates qiladi va proper role channels'ni notifies qiladi. Bu ixtironing control-path view'i.

Transactional outbox and event propagation drawing database commit va event emission'ni bitta lineage sifatida ko'rsatishi kerak. Mutating handler domain state va outbox record'ni same transaction ichida yozadi. Outbox relay keyin Kafka'ga publishes qiladi, downstream automation, notification dispatch, realtime refresh, analytics, reconciliation va exception surfaces durable stream'dan consume qiladi. Drawing event delivery database write'dan keyingi hopeful side effect emasligini ta'kidlaydi.

Realtime hub drawing warehouse, retailer, driver, payload, factory, fleet va telemetry channels'ni role-scoped live surfaces sifatida ko'rsatishi kerak. Redis Pub/Sub fan-out multi-pod delivery'ni qo'llaydi, clients esa silently freezing o'rniga reconnect yoki offline state ko'rsatishi kutiladi. Drawing live channels'ni decorative notifications emas, operational control surfaces sifatida ko'rsatishi kerak.

Auto-dispatch and planning drawing order demand, warehouse load, driver readiness, vehicle availability, H3-style spatial planning, dispatch preview, manual lock yoki override behavior, manifest generation, route execution va replenishment feedback'ni ko'rsatishi kerak. Existing auto-dispatch pipeline visual public drawing basis beradi, codebase route graph esa planning, proximity, supplier logistics

va warehouse dispatch relationships'ni qo'llab-quvvatlaydi.

Reliability control-plane drawing priority guard, rate limiting, circuit-breaker posture, Redis invalidation, idempotency, retry behavior, structured logging, metrics, role-restricted failures va stale/offline states'ni bitta safety layer sifatida ko'rsatishi kerak. Drawing faqat uptime emas, graceful degradation'ni ifodalashi kerak. Failed live channel, stale read, out-of-scope action, duplicate mutation attempt yoki overloaded low-priority request visible and bounded bo'lishi kerak.

Financial integrity drawing operational completion, payment state, ledger lineage, treasury review, reconciliation va exception handling'ni bog'lashi kerak. Logistics va finance separate domains bo'lib qoladi, lekin ularning state transitions drift qilmasligi kerak. Drawing operational facts va money movement auditable lineage baham ko'rishini ko'rsatadi.

Visual and image register drawing public architecture overview, Maglev load-balancing visual, auto-dispatch pipeline visual, reliability control-plane visual, technology-stack composites, omni-code surface artwork, role-feature diagrams, role-relations diagrams, role-surface diagrams, ecosystem workflow diagrams va Pegasus identity logo'ni group qiladi. Bu visual references PDF ichidagi formal architecture, infrastructure, system-flow, technical-stack, role-map va brand-ending pages'ni qo'llaydi, lekin document'ni implementation manual'ga aylantirmaydi.

Mathematical landscape drawing formula families'ni abstract control surfaces sifatida ko'rsatadi. Engineering and Computer Science orchestration quality and consistency'ni qamrab oladi. Radar, Positioning and Navigation location-aware confidence'ni qamrab oladi. Remote Sensing noisy observation and scene confidence'ni qamrab oladi. Physics and Mathematics stable optimization'ni qamrab oladi. General Physics and Mathematics uncertainty va automation bilan human confirmation orasidagi decision boundary'ni qamrab oladi.

Future-vision drawing assistive autonomy, predictive replenishment, exception anticipation, adaptive routing, supply-lane planning, risk-aware forecasting va operator-governed recommendations'ni ko'rsatishi kerak. Drawing recommendations confidence and authority checks bilan bounded proposals ekanini, unreviewable commands emasligini aniq ko'rsatishi kerak.

Controlled disclosure drawing disclosed va intentionally withheld material orasidagi boundary'ni ko'rsatishi kerak. Disclosed material architecture-level relationships, role behavior, visual system categories, abstract formulas va governed control loops'ni o'z ichiga oladi. Withheld material executable code, private endpoint mutation maps, schema-level contracts, deployment quantities, secrets, model constants, tuning thresholds va production wiring'ni o'z ichiga oladi. Bu boundary document'ning professional posture qismidir.

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## Batafsil figura registri

Figura 1. System infrastructure envelope. Tavsif: har bir external role surface protected routing layer'ga kirishini ko'rsatish, keyin backend handlers, live hubs, transactional storage, cache invalidation, event relay, worker automation, financial reconciliation va observability'ni alohida, ammo ulangan planes sifatida ko'rsatish. Visual Pegasus single portal emas, distributed operating system ekanini yetkazishi kerak.

Figura 2. Maglev ingress va stateless service routing. Tavsif: web, desktop, mobile va terminal clients'dan requests load distribution boundary'ga kirishini, keyin stateless backend pods va route families'ga oqishini ko'rsatish. Visual no sticky sessions, no local authority va production rotation paytida fast draining'ni ta'kidlashi kerak.

Figura 3. Route-composition contract graph. Tavsif: backend route families'ni grouped domains sifatida ko'rsatish, jumladan supplier core, supplier planning, supplier logistics, supplier operations, supplier catalog, supplier insights, retailer, driver, payload, factory, warehouse, order, delivery, payment, telemetry, webhook, treasury, proximity, simulation va infrastructure. Maqsad role and domain ownership explicit ekanini ko'rsatish.

Figura 4. Role-surface matrix. Tavsif: supplier, retailer, driver, payload, factory admin va warehouse admin'ni backend contracts, web yoki desktop surfaces, Android surfaces, iOS surfaces, terminal surfaces va realtime channels bo'ylab ko'rsatish. Visual cross-client parity'ni ko'rinarli qilishi kerak.

Figura 5. Role responsibility and handoff map. Tavsif: supplier policy factories, warehouses, drivers, retailers va payload operators'ni oziqlantirishini ko'rsatish. Factory-to-warehouse replenishment, warehouse-to-driver dispatch, payload-to-driver sealed manifest handoff, driver-to-retailer delivery proof va retailer-to-supplier demand feedback'ni ko'rsatish.

Figura 6. Full ecosystem workflow loop. Tavsif: demand planning, supplier configuration, retailer order intent, warehouse stock pressure, factory supply response, payload loading, driver route execution, retailer confirmation, exception handling, analytics, replenishment va next order cycle'ga qaytishni ko'rsatish.

Figura 7. Supplier control-plane flow. Tavsif: supplier onboarding, profile, payment configuration, catalog, pricing rules, inventory, warehouse and factory planning, fleet, manifests, dispatch, CRM, returns, analytics va treasury'ni bitta supplier-owned operating chain sifatida ko'rsatish.

Figura 8. Retailer commerce capture flow. Tavsif: supplier discovery, catalog browsing, product detail, cart, checkout, payment selection, order creation, active fulfillment, order tracking, receipt confirmation, saved cards, auto-order settings va family member management'ni ko'rsatish.

Figura 9. Driver execution spine. Tavsif: driver login, home-node scope, mission summary, route map, manifest review, QR scan, arrival validation, offload review, payment yoki cash branch, correction flow, completion va replay-safe retry behavior'ni ko'rsatish.

Figura 10. Payload loading and manifest workspace. Tavsif: truck selection, order list, selected order detail, checklist validation, mark-loaded gate, manifest exception path, seal action, dispatch success state va payload sync event propagation'ni ko'rsatish.

Figura 11. Factory replenishment and transfer loop. Tavsif: stock threshold signal, factory supply request, acceptance, production readiness, transfer manifest, loading bay, driver yoki payload handoff, warehouse receipt va supplier network planning'ga feedback'ni ko'rsatish.

Figura 12. Warehouse dispatch and replenishment loop. Tavsif: inventory posture, demand forecast, supply request creation, dispatch preview, dispatch lock acquisition, driver and vehicle availability, supply request updates, live warehouse channel va explicit restricted-state handling'ni ko'rsatish.

Figura 13. Governed state-transition path. Tavsif: request intent entering the system, identity resolution, role scope derivation, object ownership check, policy gate, idempotency replay check, version check, transactional commit, outbox emission, cache invalidation, realtime broadcast va audit lineage'ni ko'rsatish.

Figura 14. Transactional outbox relay. Tavsif: domain row va outbox row bitta transaction ichida yozilishini, undan keyin relay polling, Kafka publication, event header propagation, downstream worker consumption, notification formatting va published state mark-back'ni ko'rsatish.

Figura 15. Event cascade from order reassignment to notification. Tavsif: route yoki order reassignment, freeze-lock yoki dispatch-lock check, outbox event, Kafka relay, notification consumer, formatted message, role-scoped WebSocket broadcast va client refresh'ni ko'rsatish.

Figura 16. H3 geographic sharding and coverage planning. Tavsif: coordinates H3 cell'ga aylanishi, neighboring rings coverage'ni qo'llashi, warehouse coverage polygons cell sets'ga aylanishi, retailer demand cells'ga map qilinishi va regional data routing yoki planning decisions cell graph bilan bajarilishini ko'rsatish.

Figura 17. Warehouse load penalty curve. Tavsif: horizontal axis'da utilization va vertical axis'da recommendation penalty, load threshold ostida mild linear region va threshold ustida stronger quadratic region'ni ko'rsatish. Visual deyarli to'la warehouses geografik yaqin bo'lsa ham nega kamroq attractive bo'lishini ko'rsatishi kerak.

Figura 18. Dispatch clustering and bin-packing. Tavsif: dispatchable orders eligibility bo'yicha filtered, H3 cell va adjacent cells bo'yicha grouped, vehicle capacity bilan matched, decreasing fit bo'yicha assigned, manifest practical capacity'dan oshsa split qilinishi va route yoki manifest events sifatida emitted bo'lishini ko'rsatish.

Figura 19. Optimizer handoff and fallback. Tavsif: planning service route-solving request tayyorlashi, optimization worker'ga berishi, sequence olishi va optimizer unavailable bo'lganda deterministic bin-packing'ga fallback qilishini ko'rsatish. Visual partial dependency failure sharoitida continuity'ni ko'rsatishi kerak.

Figura 20. Human operator va AI worker o'rtasidagi freeze-lock consensus. Tavsif: manual intervention lock olishi, affected entities automation queues'dan olib tashlanishi, operator action bounded policy window ichida tugashi, audit evidence yozilishi, lock release va automation resuming'ni ko'rsatish.

Figura 21. Home-node principle. Tavsif: drivers va vehicles warehouse yoki factory bo'lishi mumkin bo'lgan home node'ga bound ekanini, driver yoki payload normal node authority'dan o'tganda inter-hub transfer manifests governed boundary bo'lishini ko'rsatish.

Figura 22. Payment settlement and double-entry ledger. Tavsif: order state, gateway authorization, capture yoki settlement, paired ledger entries, supplier wallet, retailer payment liability, platform fee account, gateway clearing account va nightly reconciliation detecting anomalies'ni ko'rsatish.

Figura 23. Payment webhook trust boundary. Tavsif: external gateway callback, signature-first verification, idempotency key derivation, typed parsing, transaction update, outbox emission, ledger effect va invalid yoki replayed inputs uchun rejection path'ni ko'rsatish.

Figura 24. Cash collection branch. Tavsif: driver cash confirmation, replay-safe mutation, payment-collected event, ledger posting, supplier credit, platform fee, route completion linkage va receipt-side confirmation'ni ko'rsatish.

Figura 25. Offline proof and correction path. Tavsif: offline delivery proof capture, local buffering, hash yoki proof envelope, later synchronization, conflict gate, replay deduplication, line-level correction, refund delta preview va canonical amendment'ni ko'rsatish.

Figura 26. WebSocket hub cross-pod relay. Tavsif: authenticated socket upgrade, room assignment by role scope, local hub fan-out, Redis Pub/Sub relay, peer-pod fan-out, heartbeat, reconnect state va cross-pod relay degraded bo'lganda fail-open behavior'ni ko'rsatish.

Figura 27. Telemetry and geofence signal loop. Tavsif: driver coordinates, route progress, fleet telemetry channel, admin yoki supplier map refresh, approach signal, nonblocking proximity event va guarded completion validation'ni ko'rsatish.

Figura 28. Priority guard and backpressure. Tavsif: incoming traffic payment yoki auth, dispatch va read tiers'ga classified bo'lishi; queue pressure rising; low-priority requests retry guidance bilan shed qilinishi; high-consequence payment yoki dispatch paths protected qolishini ko'rsatish.

Figura 29. Circuit breaker state machine. Tavsif: external provider calls closed, open va half-open states orqali o'tishi, failure thresholds, fast-fail responses, probe requests, recovery va observability metrics'ni ko'rsatish.

Figura 30. Idempotency replay gate. Tavsif: mutation request, idempotency key, body hash, first execution and stored response, same-body replay stored result qaytarishi, different-body replay conflict qaytarishi, API va webhook flows uchun separate retention windows'ni ko'rsatish.

Figura 31. Bento dashboard information mosaic. Tavsif: anchor, statistic, list, control, wide va full dashboard cells operational priority bo'yicha arranged bo'lishini ko'rsatish. Visual data density, loading skeletons, stale states va drill-down behavior'ni bog'lashi kerak, decorative dashboard cards emas.

Figura 32. Material and native UI consistency map. Tavsif: web va Android Material 3 discipline ishlatishini, iOS SwiftUI-native patterns ishlatishini, payload terminal Material-style operational density'ga amal qilishini va shared visual tokens platformalarni bir-biriga majburan o'xshatmasdan role surfaces bo'ylab oqishini ko'rsatish.

Figura 33. Supplier payment gateway configuration surface. Tavsif: providers uchun gateway cards, onboarding status, manual setup expansion, update yoki deactivate controls, supplier-scoped credential governance va downstream checkout readiness'ni ko'rsatish.

Figura 34. Retail checkout and supplier-scoped order split. Tavsif: one commerce intent checkout'ga kirishi, payment selection, supplier-scoped order creation, settlement preparation, idempotent placement va order tracking state'ni ko'rsatish.

Figura 35. Driver map as execution cockpit. Tavsif: live route map, mission markers, selected mission panel, focus controls, scan action, correction action, payment branch, geofence context va route telemetry'ni bitta operational surface ichida ko'rsatish.

Figura 36. Payload dispatch success evidence. Tavsif: sealed manifest result, active truck, secured manifest state, dispatch codes where applicable, reset to new manifest va machine-readable dispatch release'ni ko'rsatish.

Figura 37. Replenishment threshold and look-ahead loop. Tavsif: safety stock, current stock, breach detection, replenishment lock, pull matrix, predictive push, look-ahead completion, supplier notification va warehouse/factory response'ni ko'rsatish.

Figura 38. Security and scope enforcement matrix. Tavsif: JWT role, supplier scope, warehouse scope, factory scope, home-node scope, allowed route families, rejected body overrides, structured security log va operator-visible restricted state'ni ko'rsatish.

Figura 39. Visual asset production map. Tavsif: public architecture visuals, role diagrams, workflow diagrams, technology-stack composites, load-balancer artwork, reliability-control artwork, auto-dispatch artwork va Pegasus identity assets figure register bilan paired bo'lishi mumkin bo'lgan inputs sifatida ko'rsatish.

Figura 40. Controlled disclosure boundary. Tavsif: disclosed architecture-level relationships va abstract formulas bir tomonda, withheld source code, schemas, endpoint mutation maps, credentials, model constants, deployment sizes va operational thresholds boshqa tomonda bo'lishini ko'rsatish.

### Specification uchun kengaytirilgan matn mavzulari

Mavzu A. State authority va role-scoped truth. Specification har bir client surface requestor va observer ekanini, backend esa durable state transitions, role scope va audit lineage egasi ekanini tushuntirishi kerak.

Mavzu B. Role-row parity product invariant sifatida. Specification role nima uchun backend, web, desktop, Android, iOS, terminal, realtime va notification consequences'ni qamrab oluvchi product row sifatida ko'rilishini tushuntirishi kerak.

Mavzu C. Supplier portal supplier operating system sifatida. Specification catalog posture, pricing policy, onboarding, billing, gateway configuration, fleet visibility, dispatch control, manifest governance, CRM, returns, analytics, treasury va reconciliation'ni bitta supplier-owned control plane sifatida tavsiflashi kerak.

Mavzu D. Retailer commerce intent preservation. Specification discovery, cart, checkout, payment, order tracking, saved cards, family members, auto-order settings va receipt confirmation bir commercial intent'ni fulfillment and settlement bo'ylab qanday saqlashini tavsiflashi kerak.

Mavzu E. Driver execution and proof chain. Specification route assignment, mission selection, map context, QR scan, arrival validation, offload review, cash yoki digital payment branch, correction workflow va completion proof'ni bitta physical execution chain sifatida tavsiflashi kerak.

Mavzu F. Payload loading machine-readable readiness sifatida. Specification payload loading faqat manual warehouse work emas, checklist completion va manifest sealing'ni dispatch-ready evidence'ga aylantiradigan formal transition ekanini tavsiflashi kerak.

Mavzu G. Factory and warehouse node cooperation. Specification factory'ni supply generation, warehouse'ni local dispatch control sifatida, ularni node authority'ni yo'qotmasdan transfer manifests va replenishment locks bilan bog'lashini tavsiflashi kerak.

Mavzu H. Transactional outbox atomicity primitive sifatida. Specification domain mutation va durable event creation bitta commit lineage ekanini, publication relay'ga berilishi ghost entities va missing downstream notifications'ni oldini olishini tavsiflashi kerak.

Mavzu I. Cache invalidation correctness sifatida, decoration emas. Specification Redis invalidation va Pub/Sub fan-out fast read surfaces mutation'dan keyin aligned qolishini ta'minlaydigan coherence mechanism ekanini tavsiflashi kerak.

Mavzu J. Realtime channels control surfaces sifatida. Specification live role channels operational truth qismi ekanini, reconnection, offline state, stale state va relay degradation paytida fail-open local delivery'ni qamrab olishini tavsiflashi kerak.

Mavzu K. Human override with automation standoff. Specification dispatch locks va freeze-lock behavior automation'dan human operator'ga bounded authority transfer bo'lishini, auditability va re-engagement bilan tavsiflashi kerak.

Mavzu L. Cell-based planning bilan spatial reasoning. Specification geospatial routing, coverage, warehouse selection, retailer demand mapping, dispatch clustering va regional planning raw coordinate scanning emas, cell-based logic ekanini tavsiflashi kerak.

Mavzu M. Financial reconciliation fulfillment consequence sifatida. Specification ledger entries, payment settlement, cash collection, gateway callbacks, treasury review va anomaly detection fulfillment state bilan bog'liqligini tavsiflashi kerak.

Mavzu N. Partial failure sharoitida reliability. Specification priority guard, circuit breakers, idempotency, retry behavior, stale reads, offline UI states va fallback route planning unified reliability posture ekanini tavsiflashi kerak.

Mavzu O. Future assistive autonomy. Specification predictive replenishment, risk-aware routing, adaptive supply lanes, exception anticipation va confidence-bounded recommendations operator authority preserved holda ishlashini tavsiflashi kerak.

### Kengaytirilgan formula va logic mavzulari

Formula mavzusi 1. Warehouse load penalty. Tavsif: recommendation cost distance multiplied by load penalty sifatida beriladi, penalty normal utilization ostida sekin, warehouse saturation'ga yaqinlashganda esa agressivroq o'sadi.

Formula mavzusi 2. Dispatch fit score. Tavsif: distance to first stop, vehicle capacity utilization, driver availability, route cohesion, payment readiness va freeze-lock exclusion bitta assignment score'ga birlashtiriladi.

Formula mavzusi 3. H3 coverage confidence. Tavsif: service confidence target cell membership, neighboring cell coverage, warehouse capacity, travel distance va same region'dagi recent dispatch success'dan baholanadi.

Formula mavzusi 4. Route stability loss. Tavsif: local distance'ni yaxshilaydigan, lekin excessive resequencing, stale instructions, policy violation yoki driver confusion kiritadigan route plans penalize qilinadi.

Formula mavzusi 5. Idempotency replay result. Tavsif: replay behavior key, body hash, stored response, endpoint scope va retention window function'i sifatida model qilinadi.

Formula mavzusi 6. Outbox delivery risk. Tavsif: downstream risk committed-but-unpublished age, relay retry count, topic health va consumer lag function'i sifatida beriladi.

Formula mavzusi 7. Realtime freshness score. Tavsif: last update age, socket state, reconnect attempts, role room identity va local cache age visible freshness indicator'ga birlashtiriladi.

Formula mavzusi 8. Ledger balance invariant. Tavsif: money correctness credits and debits per currency and reconciliation window zero bo'lishi kerakligi, anomalies yashirilmasdan raised bo'lishi sifatida beriladi.

Formula mavzusi 9. Priority shedding boundary. Tavsif: request shedding queue depth, endpoint tier, actor rate va retry-after policy threshold function'i sifatida beriladi.

Formula mavzusi 10. Human confirmation entropy. Tavsif: candidate actions high ambiguity, high financial consequence, high route disruption yoki insufficient observational confidence bo'lganda human review talab qilinadi.

## BATAFSIL TAVSIF

### Texnik disclosure

Texnik jihatdan Pegasus contract-first logistics control plane hisoblanadi. Backend durable truth va state transitions egasi. Clients local truth ixtiro qilmaydi. Ular request, display, confirm va recover qiladi. Har bir role product surface'ga ega, va har bir role surface uni feed qiladigan backend contract bilan aligned bo'ladi. Bu muhim, chunki driver app route'ni warehouse portal'dan boshqacha tushunsa, bu kichik UX bug emas. Bu physical operations risk.

System additive compatibility'ga tayanadi, casual renaming'ga emas. Route, event yoki payload shape o'zgarganda older clients coordinated migration bo'lmasa ishlashda davom etishi kerak. Shu sababli architecture stable DTOs, role-row parity, explicit compatibility aliases va route-composition packages ownership'ni afzal ko'radi. Oddiy aytganda, Pegasus bir app jimgina oldinga o'tib, boshqa surfaces hech narsa bo'lmagandek ko'rinishidan qochadi.

## Notexnik disclosure

Notexnik jihatdan Pegasus trust atrofida qurilgan. Supplier warehouse bir xil business reality ko'rishiga ishonadi. Warehouse driver executable work olganiga, stale theory emasligiga ishonadi. Retailer confirmed demand va receipt payment and fulfillment'ga to'g'ri ta'sir qilishiga ishonadi. Payload team sealed manifest operational ma'noga ega ekaniga ishonadi. Leadership exception bo'lganda system kim act qilganini, qaysi authority ostida va nima o'zgarganini tushuntira olishiga ishonadi.

Ko'p software description aynan shu qismni tashlab ketadi. Ixtiro dispatch, maps, payments, dashboards va apps borligida emas. Bular expected. Ixtiro bu surfaces bir polished portal yonidagi zaif side channels emas, bitta controlled operating environment sifatida ishlashida.

## Infra, arxitektura, logika, maqsad, g'oya va oqim

Infrastructure planes'ga bo'lingan, shunda system scale qilganda har bir request database fight'ga aylanmaydi. Request traffic backend handlers'ga yetishdan oldin routing and protection layer'dan o'tadi. Backend handlers actor identity, scope va policy'ni resolve qiladi. Transactional data stores durable truth'ni saqlaydi. Cache va Pub/Sub invalidation read speed'ni saqlaydi, lekin stale correctness'ni normal deb qabul qilmaydi. Eventing state changes'ni workers va role surfaces'ga olib boradi. Live channels operational screens'ni fresh saqlaydi. Observability logs, events va operator-visible behavior'ni birga bog'laydi.

Architecture role-row oriented. Role faqat bitta client emas. Role web, desktop, mobile, terminal, backend, event va notification consequences'ga ega business actor. Supplier, driver, retailer, payload, factory va warehouse surfaces har biri o'z contract row'ga ega. Backend role capability'ni kengaytirganda corresponding role clients shape'ni tushunishi kerak yoki parity bo'lmaguncha system capability'ni hide qilishi kerak.

Logic guarded transition model'ga amal qiladi. Request permission emas, intent sifatida boshlanadi. System actor'ni authentication'dan, operational scope'ni claims and node relationships'dan resolve qiladi, target object requested state'ga o'ta olishini evaluate qiladi, stale yoki replayed attempts'ni rejects qiladi, state change'ni commits qiladi, durable event'ni emits qiladi, affected read models'ni invalidates qiladi va relevant role channels'ni notifies qiladi. Bu ordinary CRUD'dan qattiqroq. Logistics physical. Mistaken state change truck yuborishi, manifest unlock qilishi yoki liability shift qilishi mumkin.

Purpose speed va correctness'ni birga ushlashdir. Silent state drift yaratadigan fast software logistics'da foydasiz. Constant manual reconciliation talab qiladigan correct software operationally viable emas. Pegasus system default bo'yicha optimize qilishini, ammo ground truth default plan'ga mos kelmaganda humans'ga bounded control berishini ta'minlash uchun designed.

Flow ortidagi idea shuki, har bir operational object lineage olib yuradi. Order shunchaki complete bo'lmaydi. U accepted, assigned, loaded, transported, arrived, verified, completed va reconciled bo'ladi, chain of evidence orqali. Manifest shunchaki sealed bo'lmaydi. U assembled, checked, started, sealed, dispatched va reality deviates bo'lsa exception-handled bo'ladi. Payment shunchaki settled bo'lmaydi. U fulfillment state bilan tied va append-only accounting logic orqali reconciled bo'ladi.

## Role behavior

Supplier role commercial and operational owner hisoblanadi. Current product doctrine ichida Admin Portal Supplier Portal hisoblanadi. Supplier catalog posture, pricing intent, operational policy, warehouse and factory planning, fleet visibility, analytics, billing setup va exception governance'ga egalik qiladi. Supplier role generic platform administrator degani emas. U o'z logistics network'i uchun responsible business operator degani.

Factory admin role production-side readiness va supply generation'ga egalik qiladi. Factory role transfers tayyorlashi, replenishment demand'ga javob berishi, staff and fleet resources'ni coordinate qilishi va production-side manifest state'ni warehouse needs bilan coherent saqlashi mumkin. Uning authority node-scoped, chunki factory operator UI'da field borligi uchun boshqa node'ni mutate qilmasligi kerak.

Warehouse admin role local fulfillment control'ga egalik qiladi. Warehouse stock posture, staff availability, dispatch locks, supply requests, vehicle availability, driver readiness va live operational state'ni kuzatadi. Warehouse role automation va human override eng ko'rinarli uchrashadigan joy. Warehouse operator intervene qilishi mumkin, lekin intervention conflicting automation'ni pause qilishi va audit trail qoldirishi kerak.

Driver role physical execution'ga egalik qiladi. Driver route work oladi, arrival validates qiladi, delivery steps'ni confirms qiladi, kerak bo'lganda missing items reports qiladi va high-consequence actions'ni replay-safe requests orqali completes qiladi. Driver role home node model'ga scoped, shunda execution authority loosely trusted request body emas, driver identity'ga ergashadi.

Retailer role demand, receipt va payment-facing confirmation'ga egalik qiladi. Retailer orders places yoki modifies qiladi, allowed bo'lganda AI-assisted suggestions'ga responds qiladi, fulfillment state'ni follows qiladi, card and cash payment flows'ni handles qiladi va receipt-side reality'ni confirms qiladi. Retailer identity terms bo'yicha supplier ownership tashqarisida, ammo uning order and payment events supplier operations bilan reconcile bo'lishi kerak.

Payload role loading va manifest integrity'ga egalik qiladi. Payload surfaces trucks, orders, seals, missing-item reporting, reassignment recommendations va payloader live sync atrofida ishlaydi. Bu role deliberately separate, chunki loading errors keyin expensive bo'ladi. System payload operators'ga first-class workflow beradi, loading'ni generic warehouse screen ichiga yashirmaydi.

### Texnik va notexnik qiymat

Texnik qiymat shundaki, Pegasus distributed logistics'ga shared transition grammar beradi. Role-scoped authorization spoofing risk'ni kamaytiradi. Transactional persistence ghost states'ni kamaytiradi. Outbox eventing missed downstream updates'ni kamaytiradi. Cache invalidation stale read traps'ni kamaytiradi. WebSocket channels live operations blind spots'ni kamaytiradi. Idempotency duplicate high-consequence mutations'ni kamaytiradi. Version-aware behavior stale replay damage'ni kamaytiradi. Bu pieces decorative emas. Ular logistics systems orasidagi edge'larda buzilgani uchun mavjud.

Notexnik qiymat shundaki, odamlar operating picture'ga ishonishi mumkin. Supplier risk qayerda building ekanini ko'radi. Warehouse dispatch nega changed bo'lganini tushuntira oladi. Driver action'ni duplicate qilmasdan retry qila oladi. Retailer internal logistics'ni tushunmasdan fulfillment'ni track qiladi. Payload operator loading truth'ni early report qiladi. Finance system ortidan cleanup qilish o'rniga same lineage'dan reconcile qiladi.

### Engineering and Computer Science formula landscape

Bu landscape orchestration quality'ni model qiladi. Formula PDF export'dan omon o'tishi va math renderer'siz o'qilishi uchun plain notation'da yozilgan.

```
Q_ops = alpha_valid * valid_transition_rate + alpha_sync * role_sync_score - alpha_conflict *
                      conflict_rate - alpha_latency * propagation_delay
```

Formula system goal'ni engineering terms'da tasvirlaydi. Valid transitions oshishi kerak. Role synchronization oshishi kerak. Conflict rate va propagation delay tushishi kerak. Exact weights ataylab disclosed qilinmaydi, chunki ular implementation-sensitive. Patent-level idea correctness, role agreement, conflict va propagation delay'ni bitta control surface sifatida ko'radigan composite orchestration score ishlatishdir.

### Radar, Positioning and Navigation formula landscape

Bu landscape location-aware confidence'ni model qiladi. Logistics decisions ko'pincha actor, vehicle, warehouse, retailer yoki route aslida qayerda ekaniga bog'liq, lekin location readings imperfect. Shuning uchun system position'ni blind coordinate lookup emas, confidence problem sifatida ko'radi.

```
p_hat(t) = argmin_over_p SUM[k in S(t)] w_k * residual_score(signal_k, p, time_lag_k)
```

Formula chosen position estimate available signals va time lag bo'yicha weighted residual error'ni minimallashtiradigan candidate position ekanini bildiradi. Ixtiro bitta specific positioning vendor talab qilmaydi. U location-sensitive actions completion, dispatch, arrival yoki exception state'ga ta'sir qilishdan oldin confidence model orqali evaluated bo'lishini talab qiladi.

### Remote Sensing formula landscape

Bu landscape operational scene confidence'ni model qiladi. Logistics'da observed scene incomplete bo'lishi mumkin. Signal delayed, occluded, low quality yoki boshqa source bilan inconsistent bo'lishi mumkin. Platform therefore scene state'ni simple yes/no emas, confidence score sifatida ko'radi.

```
C_scene = beta_signal * signal_quality + beta_coherence * cross_signal_agreement - beta_occlusion *
                        occlusion_penalty - beta_staleness * data_age
```

Formula warehouse, route, driver yoki manifest state nega barcha observation conditions'da bir xil trusted bo'lmasligini ko'rsatadi. High signal quality va cross-signal agreement confidence'ni oshiradi. Occlusion va stale data confidence'ni kamaytiradi. Protected concept scene confidence'ni automation va escalation gate sifatida ishlatishdir.

### Physics and Mathematics formula landscape

Bu landscape stable optimization'ni model qiladi. Routing, replenishment, dispatch va balancing systems har bir small change ortidan quvsa unstable bo'lishi mumkin. Pegasus regularized objective g'oyasidan foydalanadi. System current reality'ga yaxshi fit izlaydi, instability va policy violation'ni penalize qiladi.

```
Loss(x) = lambda_fit * norm2(x - x_hat)^2 + lambda_smooth * norm1(gradient(x)) + lambda_policy *
                                        policy_penalty(x)
```

Bu avval yomon rendered bo'lgan formula uchun corrected format. U backslash mathcal va backslash lambda kabi raw LaTeX commands'dan qochadi. So'z bilan aytganda, system candidate plan'ni observed plan bilan taqqoslaydi, unnecessary jagged changes'ni penalize qiladi va plan business yoki safety constraints'ni buzsa policy penalty qo'shadi. Constants disclosed qilinmaydi, chunki ular operational tuning'ni encode qiladi.

### General Physics and Mathematics formula landscape

Bu landscape uncertainty'ni model qiladi. Automation uncertainty low bo'lganda act qilishi, uncertainty high bo'lganda human confirmation so'rashi kerak. System decision ambiguity'ni entropy sifatida ko'rishi mumkin.

```
Entropy(P) = - SUM[i] p_i * log(p_i), with SUM[i] p_i = 1 and 0 <= p_i <= 1
```

Low entropy system clear best action'ga ega ekanini bildiradi. High entropy system multiple plausible interpretations ko'rayotganini bildiradi. Non-technical translation oddiy. System confident bo'lmasa, pretending'ni to'xtatib right operator'dan so'rashi kerak.

### Future vision features

Pegasus'ning future version'i operators'ni itaradigan black box bo'lmasligi kerak. Kuchliroq yo'l assistive autonomy. System dispatch batches, replenishment timing, route changes, supply-lane shifts, warehouse territory changes, payment exception handling va staffing adjustments tavsiya qilishi mumkin, lekin role authority va auditability saqlanadi.

Next generation features risk ticket'ga aylanishidan oldin visible qilishi kerak. Warehouse drivers kutishidan oldin likely dispatch bottleneck'ni ko'rishi kerak. Supplier inventory collapses bo'lishidan oldin forecast drift'ni ko'rishi kerak. Factory warehouse escalating boshlashidan oldin replenishment pressure'ni ko'rishi kerak. Retailer internal state'ni decode qilmasdan fulfillment confidence'ni ko'rishi kerak. System healthy bo'lganda quiet, nosog'lom bo'lganda precise bo'lishi kerak.

Future vision ortidagi patent-relevant idea prediction emas. Prediction alone cheap. Kuchliroq idea prediction with governed actuation. Recommendation faqat system role scope, confidence level, expected consequence, rollback path va qoladigan audit evidence'ni explain qila olganda foydali bo'ladi.

### Qo'shimcha professional maydonlar

Novelty posture: Pegasus role-row contract integrity, transactional event lineage, geospatial confidence, manifest governance, replay safety va financial reconciliation'ni bitta logistics control plane ichida birlashtiradi. Novelty eng kuchli joy har bir mechanism alohida turganida emas, ular interaction qilganida.

Industrial applicability: System supplier-led distribution, factory-to-warehouse replenishment, warehouse dispatch, direct-to-retailer fulfillment, payload loading, driver execution, route monitoring, payment reconciliation va exception handling'ga qo'llanadi. Same control model regulated delivery, cold-chain logistics, high-value goods, route-sensitive fulfillment va multi-node regional supply networks'ga moslashtirilishi mumkin.

Reliability posture: System visible ways'da degrade bo'lishi uchun designed. Dropped live channel reconnect qilishi yoki offline state ko'rsatishi kerak. Stale view trusted bo'lish o'rniga labeled bo'lishi kerak. Retry mutation duplicate qilish o'rniga safely replay bo'lishi kerak. Manual override automation bilan race qilish o'rniga affected entity'ni lock qilishi kerak.

Security posture: Scope client supplied convenient field'dan emas, authenticated identity va node relationship'dan kelishi kerak. Mutating actions replay-safe, audit-backed va role-bound bo'lishi kerak. External integrations body parsing yoki state mutation'dan oldin verified bo'lishi kerak. Bu brief secret material, signing details, private endpoints yoki production topology'ni oshkor qilmaydi.

Commercial posture: Platform operational density kerak bo'lgan, lekin control'ni yo'qotmasligi zarur biznesni qo'llab-quvvatlaydi. U supplier'ga network level'da operate qilish, node operators'ga locally act qilish, drivers'ga guessing'siz execute qilish, retailers'ga fulfillment view'ga trust qilish va finance'ga same lineage'dan reconcile qilish imkonini beradi.

Reverse-engineering posture: Bu document'ning o'zidan direct reproduction ataylab impractical. Description executable source, database schema details, private constants, full endpoint maps, model tuning values, credential flows, infrastructure sizing va failure-mode thresholds'ni omit qiladi. Document invention'ni review uchun yetarlicha explain qiladi, lekin working system'ni clone qilish uchun yetarli emas.

## Professional yakuniy tavsif

Pegasus'ni multiple human roles va software surfaces bo'ylab yoyilgan governed logistics state machine sifatida tushunish eng to'g'ri. U consistency'ni enforce qilish uchun yetarlicha technical, logistics teams aslida qanday ishlashiga mos kelish uchun yetarlicha practical va automation quietly take authority it should not have qilmasligi uchun yetarlicha cautious. System truth shared, authority scoped, transitions auditable va future automation operation'ni yuritadigan odamlarga answerable bo'lganda yutadi.

PEGASUS

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